

College Allocation Database System

Team Members:

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Problem Selected:

College Allocation Database System

Problem Definition:

The aim of the project is to develop a clean and structured database management system for college allocation. This system would be used by Students and College Allocation Authorities to manage student details and allocate college based on student preferences. Every student will have details like Name, Email, Mobile Number, Category, General Rank, Category Rank(if applicable). The Database will also store the data of College details such as College Name, Departments, Number of seats per Department for each Category. The Categories include OPEN, OPEN-PWD, GEN-EWS, GEN-EWS-PWD, SC, SC-PWD, ST, ST-PWD, OBC-NCL, OBC-NCL-PWD and Gender based reservation for each of the above categories.

Assumptions:

- We are considering 10 IITs with 5 Departments per college for testing and demo purposes along with a dataset of ~7000 students.
- We assume that student details (which include above mentioned details) are given by the Examination Authorities(like NTA).
- The testing data for the demo is taken for a few IITs, taken from the official JOSAA website and is directly taken into the database but for real life use cases administrators can add data of colleges through the website.
- Student Dataset will be generated using python script for testing and demo purposes.
- The allocation process will be of 3 Rounds.
- We are not considering fee payment for confirmation of the seat.

- We are limiting the maximum number of preferences to be 20.

Student Pages:

Upon student login they can view their details and there will be dedicated pages to

- Fill in the college preferences and this page will also display allocated preference after each round. Students will be able to filter from the given list of colleges and add them to their preference list and then arrange them prior to submission. Once the deadline to edit the preference list is crossed the preferences of the student cannot be changed any more.
- Also in the page mentioned above, students will have the option to freeze, float and withdraw from the counselling process after each and every round.
- Display round based opening and closing ranks when searched for a particular input(which includes round number, college, department, category).
- Display total number of seats when searched for a particular college and department.

Admin Pages:

Admin will be able to login to the application with credentials provided by the developers, from which admin would be able to

- Start the process of seat allocation for all the students.
- View the details related to round based allocation of seats based on college, category.
- Admin would also be able to view details related to all the students.

Tentative Plan for Implementation:

1. Fetching information and data related to allocation system:

This process would include collecting and understanding details related to how the seat allocation system would work in the real world i.e from sources like JOSAA official documentation to understand the process of allocation. And try to implement something similar which would follow official guidelines. This step would also include collecting details related to the number of seats present in the selected colleges for testing purposes. Student data would be generated by usage of another python script to generate dummy data for the application testing and demo purposes.

2. Database implementation:

We would store the details related to students, colleges, preferences and details related to the round based opening and closing ranks for every college and department (this would also include category based opening and closing ranks). Next step in the implementation would be writing an algorithm for the seat allocation for students based on their merit, category based ranks based on the preferences they have provided prior to the start of the allocation rounds.

3. Logic behind seat allotment:

The process of seat allotment would start by arranging all the students based on open category rank. Then we start from the first student and take his first preference and check if he would get his preference by his open rank if yes we allot the seat and move on to next student else we check with his category rank (if applicable), in any way if the student is unable to get his seat for that preference, we move on to the next preference and keep doing until all his preferences are checked, if none of them are suitable we do not allocate the seat to the student and save the same in the database. The above mentioned process repeats for every student in the order of the ranks.

4. Definitions of freeze, float and withdraw:

After every round students will have the option to freeze, float and withdraw from the allocation process.

- **Freeze:** If this option is selected by the student then he/she will be removed from the allocation process for the next rounds and this means that the student has accepted the allocated seat.
 - **Float:** If this option is selected by the student then it means that the student has accepted the seat and is willing to participate in the next rounds for the possibility of getting a higher preference.
 - **Withdraw:** If this option is selected then it means that the student has decided to go out of the allocation process and has rejected the allotted seat, declaring that he will not participate in future rounds, nor will he/she be able to get the seat allocated in the previous rounds.
5. We will also be storing the details of opening and closing rank for every college branch (this includes category based opening and closing ranks as well) in the database so that users can view the details of colleges of every round.

6. The users can also view the seat matrix i.e details related to every college, department and category wise number of seats present.

7. Tech Stacks:

For the frontend we will be using NextJS and for the API endpoints, logic handling we will be using NextJS inbuilt API page router structure. For database management we will be using MySQL.